DIGITAL LOGIC DESIGN ASSIGNMENT

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Course: Digital Logic Design

Section 1: Number System Conversions

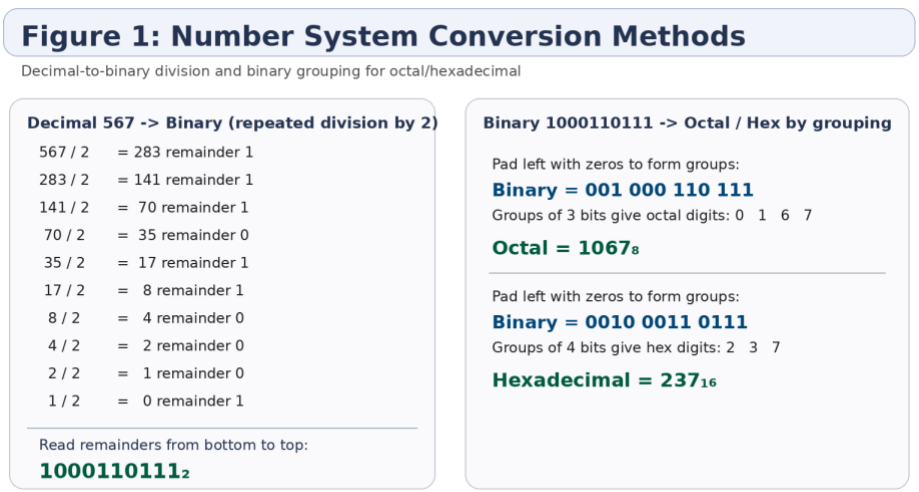
1. Decimal 567 Conversion:

Binary: 1000110111
Octal: 1067
Hexadecimal: 237

2. Binary 10110000111001 Conversion:

Decimal: 11321
Octal: 26071
Hexadecimal: 2C39

Figure 1: Number System Conversion Methods



Section 2: Conceptual Questions

Digital Logic Design (DLD) is the foundation of digital systems used in computers and electronics.

LSB is the least significant bit, while MSB is the most significant bit in a binary number.

1's complement is obtained by inverting bits. 2's complement is obtained by adding 1 to the 1's complement.

Figure 2: MSB and LSB Representation

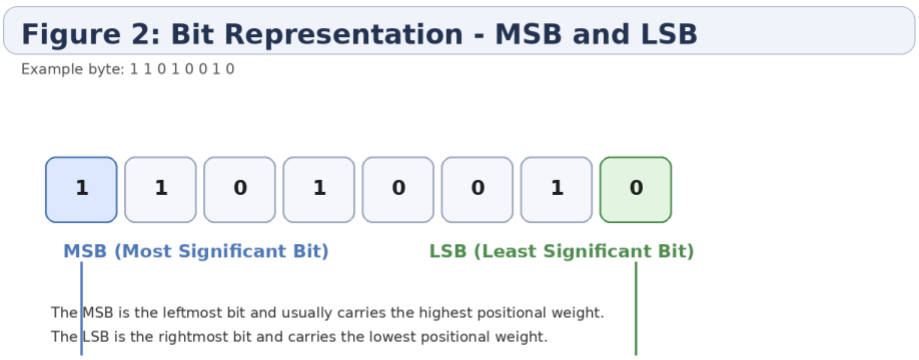
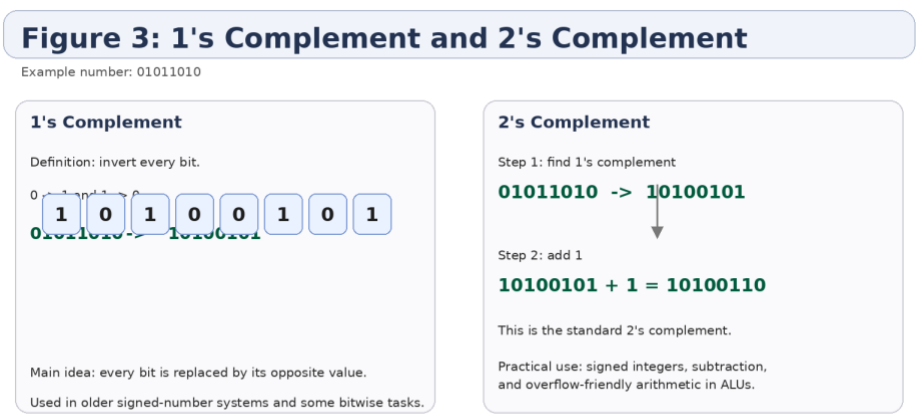


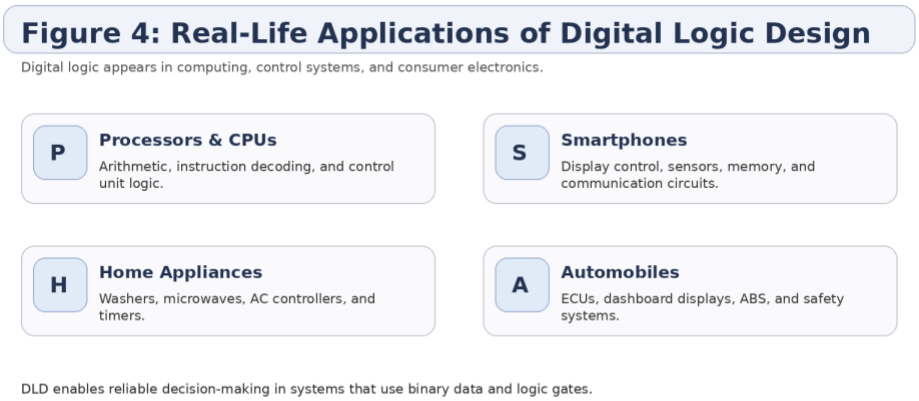
Figure 3: Complement Methods



Section 3: Applications of DLD

DLD is used in processors, embedded systems, and digital devices like smartphones and computers.

Figure 4: Real-life Applications



References

Morris Mano, Digital Design.
Wikipedia.
GeeksforGeeks.